

UGED1111 Logic

Natural Deduction I

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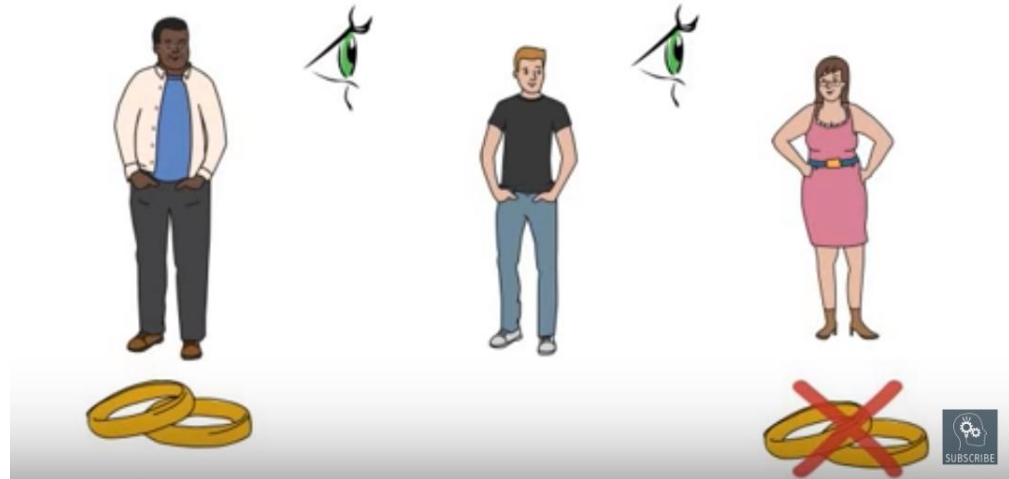
Scenario

Is a married person looking at an unmarried person?

A: Yes

B: No

C: Not enough
information



Scenario

1. 當小蘭在廚房或小東不在廚房，則廚師會在廚房。
 2. 只要小東在廚房，則小北也在廚房。。
 3. 並不是小北或小七在廚房。
 4. 廚師在廚房。
-

1. $(P \vee \sim R) \supset S$
2. $R \supset Q$
3. $\sim (Q \vee M) // S$

P:小蘭在廚房。
Q:小北在廚房。
R:小東在廚房。
S:廚師在廚房。
M: 小七在廚房



Checking validity

(P	v	$\sim R) \supset S / R \supset Q / \sim (Q \vee M) // S$

Valid argument.

But is this all that we want to know???

We also want to know how he deduces the conclusion from the premises!

Natural Deduction



- ❑ Natural deduction is a method of deriving the conclusion of a valid argument from the premises through a series of discrete steps.
- ❑ This method aims to demonstrate the reasoning process of an argument (how we prove the conclusion from the premises).
- ❑ However, it does not work for invalid arguments.

18 Rules of Inference

There are 18 rules of inference.

□ **8 Rules of Implication**

+

□ **10 Rules of Replacement**

*****All 18 rules will be given in the exam. So you don't need to bother memorizing them.**

Rules of implication I

1. Modus Ponens (MP)

$p \supset q$

p

q

2. Modus Tollens (MT)

$p \supset q$

$\sim q$

$\sim p$

3. Hypothetical Syllogism (HS)

$p \supset q$

$q \supset s$

$p \supset s$

4. Disjunctive Syllogism (DS)

$p \vee q$

$\sim p$

q

Rules of implication II

5. Constructive Dilemma (CD)

$(p \supset q) \cdot (r \supset s)$

$p \vee r$

$q \vee s$

6. Simplification (Simp)

$p \cdot q$

p

7. Conjunction (Conj)

p

q

$p \cdot q$

8. Addition (Add)

p

$p \vee q$

1. Modus Ponens (MP)

Given a conditional statement , we can deduce its consequent by asserting its antecedent.

$$p \supset q$$
$$p$$

$$q$$

1. If you get GPA 4 , you will get the scholarship ◦

2. You get GPA 4 ◦

3. You will get the scholarship ◦

2. Modus Tollens (MT)

Given a conditional statement and the negation of its consequent, then we can deduce the negation of its antecedent.

$$p \supset q$$
$$\sim q$$

$$\sim p$$

1. If the temperature is higher than 30 degree Celsius , then the school will turn on the air-conditioner .
2. The school did not turn on the air-conditioner today .

3. It is less than 30 degree Celsius today.

3. Hypothetical Syllogism (HS)

Given two conditional statements with a common formula as transition then we can deduce a new conditional without that common formula.

$$p \supset q$$
$$q \supset s$$

$$p \supset s$$

1. 如果我收到利息，則母親幫我保存利息。
 2. 如果母親幫我保存利息，則利息變成母親存款。
-
3. 如果我收到利息，則利息變成母親存款。

4. Disjunctive Syllogism (DS)

Given a disjunctive statement and the negation of its left-hand disjunct, we can deduce the right-handed disjunct.

$$p \vee q$$
$$\sim p$$

$$q$$

1. 小明是帥哥或者有才華的。

2. 小明不是帥哥。

3. 小明是有才華的。

Discussion 1a

Construct a formal proof of validity for the following arguments by Natural Deduction.

1. $P \supset Q$

2. $Q \supset S$

3. $\sim S \quad // \sim P$

4. $P \supset S \quad 1, 2 \text{ HS}$

5. $\sim P \quad 3, 4 \text{ MT}$

4. $\sim Q \quad 2, 3 \text{ MT}$

5. $\sim P \quad 1, 4 \text{ MT}$

Discussion 1b

Construct a formal proof of validity for the following arguments by Natural Deduction.

1. $P \vee S$
2. $S \supset Q$
3. $\sim P \quad //Q$
4. $S \quad \quad \quad 1,3 \text{ DS}$
5. $Q \quad \quad \quad 2,4 \text{ MP}$

5. Constructive Dilemma (CD)

Given a conjunction of two conditional statements.
If either of the antecedents is true, then either of
the consequents is true.

$(p \supset q) \cdot (r \supset s)$

$p \vee r$

$q \vee s$

1. 如果下雨則賣水鞋，並且如果天晴則賣傘。

2. 下雨或天晴。

3. 賣水鞋或賣傘。

6. Simplification (Simp)

Given a conjunction of two statement, then we can deduce the left-sided conjunct.

$$\frac{p \cdot q}{p}$$

1. 殺老師反應很快，而且他很聰明。

2. 殺老師反應很快。



7. Conjunction (Conj)

Given that two propositions are true separately,
then their conjunction are also true.

p

1. I have a pen.

q

2. I have an apple.

$p \cdot q$

3. I have a pen and a apple.

AR!Apple pen!



8. Addition (Add)

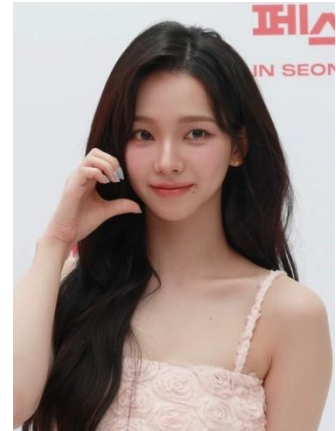
Given a proposition is true, its disjunction with any proposition is also true.

p

$p \vee q$

1. Karina很漂亮。

2. Karina很漂亮，或者Karina很可愛。



Discussion 2

Construct a formal proof of validity for the following arguments by Natural Deduction.

1. $P \cdot M$
2. $P \supset Q$
3. $R \supset S // Q \vee S$
4. $(P \supset Q) \cdot (R \supset S)$ 2,3 Conj
5. P 1, Simp
6. $P \vee R$ 5, Add
7. $Q \vee S$ 4,6 CD

Exercise 1

1. $\sim C \supset (A \supset C)$

2. $\sim C$ $// \sim A$

3. $A \supset C$ 1,2, MP

4. $\sim A$ 2,3, MT

Exercise 2

1. $\sim S \supset D$
2. $\sim S \vee (\sim D \supset K)$
3. $\sim D$ // K
4. $\sim\sim S$ 1,3, MT
5. $\sim D \supset K$ 2,4, DS
6. K 3,5, MP

Invalid inference

Rules of implication cannot be applied on a part of compound statements.

1. $P \supset Q$
2. $(P \vee R) \supset Q$ 1, Add

1. $(P \cdot R) \supset Q$
2. $P \supset Q$ 1, Simp

1. $P \supset Q$
2. $(P \supset Q) \vee R$ 1, Add

1. $(P \supset Q) \cdot R$
2. $P \supset Q$ 1, Simp

Improper use of rules

Even though the following inferences are valid, they are improper.

1. P
2. $Q \vee P$ 1, Add

1. $P \cdot R$
2. R 1, Simp

1. P
2. $P \vee Q$ 1, Add

1. $P \cdot R$
2. P 1, Simp

Rules of replacement I

9. DeMorgan's Rule (DM)

$$\sim(p \cdot q) :: \sim p \vee \sim q$$

$$\sim(p \vee q) :: \sim p \cdot \sim q$$

10. Commutativity (Com)

$$p \vee q :: q \vee p$$

$$p \cdot q :: q \cdot p$$

11. Association (Assoc)

$$p \vee (q \vee r) :: (p \vee q) \vee r$$

$$p \cdot (q \cdot s) :: (p \cdot q) \cdot s$$

12. Distribution (Dist)

$$p \cdot (q \vee r) :: (p \cdot q) \vee (p \cdot r)$$

$$p \vee (q \cdot s) :: (p \vee q) \cdot (p \vee s)$$

13. Double Negation (DN)

$$p :: \sim\sim p$$

9. DeMorgan's Rule (DM)

“Not both p and q” is logically equivalent to “Not p or not q” ;
and

“Not either p or q” is logically equivalent to “Not p and not q.”

$$\sim (p \cdot q) :: \sim p \vee \sim q$$

並不是大嘅鬼和「任何人」都有頭髮。
大嘅鬼沒有頭髮或「任何人」沒有頭髮。

$$\sim (p \vee q) :: \sim p \cdot \sim q$$

並不是大嘅鬼或「任何人」有穿衣服。
大嘅鬼沒有穿衣服而且「任何人」沒有穿衣服。



10. Commutativity (Com):

The truth value of a conjunction or disjunction is unaffected by the order in which the components are listed.

$$p \vee q :: q \vee p$$

今天是晴天，或者今天是陰天。
今天是陰天，或者今天是晴天。

$$p \cdot q :: q \cdot p$$

熊本熊和鬆弛熊來中大。
鬆弛熊和熊本熊來中大。

11. Association (Assoc)

When the same operator is used through out a conjunctive or disjunctive statement, the placement of brackets will not alter the truth value of the whole statement.

$$p \vee (q \vee r) :: (p \vee q) \vee r$$

今天是晴天，或者今天是陰天或雨天。
今天是晴天或陰天，或者今天是雨天。

$$p \cdot (q \cdot s) :: (p \cdot q) \cdot s$$

北極熊來中大，而且黑熊和啡熊來中大。
北極熊和黑熊來中大，而且啡熊來中大。

12. Distribution (Dist):

When a proposition is conjoined to a disjunctive statement in bracket or disjoined to a conjunctive statement in bracket, the rule allows us to put that proposition together with each of the components inside the bracket, and also to go in the reverse direction.

$$p \cdot (q \vee r) :: (p \cdot q) \vee (p \cdot r)$$

北極熊來中大，並且黑熊或啡熊來中大。
北極熊和黑熊來中大，或者北極熊和啡熊來中大。

$$p \vee (q \cdot s) :: (p \vee q) \cdot (p \vee s)$$

Similar technique you have used before

$$2 \times (4+3) = (2 \times 4) + (2 \times 3)$$

$$m \times (n+y) = mn + my$$

13. Double Negation (DN)

If pairs of tildes immediately adjacent to one another, deleting or introducing them cannot change the truth value of the whole statement.

$p :: \sim\sim p$

我沒有不上課。
我有上課。

負負得正?!

Discussion 3

1. $\sim (\sim P \vee Q)$
2. $P \supset S$ //S
3. $\sim\sim P \cdot \sim Q$ 1, DM
4. $P \cdot \sim Q$ 3, DN
5. P 4, Simp
6. S 2,5 MP

Discussion 4

1. $D \bullet (E \vee F)$
2. $\sim D \vee \sim F \quad // \quad D \bullet E$
3. $(D \bullet E) \vee (D \bullet F) \quad 1, \text{Dist}$
4. $\sim (D \bullet F) \quad 2, \text{DM}$
5. $(D \bullet F) \vee (D \bullet E) \quad 3, \text{Com}$
6. $D \bullet E \quad 4, 5, \text{DS}$

Exercise 3

1. $(K \cdot H) \vee (K \cdot L)$
2. $\sim L$ // H
3. $\sim L \vee \sim K$ 2, Add
4. $\sim (L \cdot K)$ 3, DM,
5. $\sim (K \cdot L)$ 4, Com
6. $(K \cdot L) \vee (K \cdot H)$ 1, Com
7. $K \cdot H$ 5,6, DS
8. $H \cdot K$ 7, Com
9. H 8, Simp

Exercise 4

- | | | |
|----|-------------------------------|--------------|
| 1. | $S \vee (P \cdot Q)$ | |
| 2. | $(S \vee Q) \supset \sim R$ | |
| 3. | $(R \vee P) \vee Q$ | / $P \vee Q$ |
| 4. | $(S \vee P) \cdot (S \vee Q)$ | 1, Dist |
| 5. | $(S \vee Q) \cdot (S \vee P)$ | 4, Com |
| 6. | $S \vee Q$ | 5, Simp |
| 7. | $\sim R$ | 2,6, MP |
| 8. | $R \vee (P \vee Q)$ | 3, Asso |
| 9. | $P \vee Q$ | 7,8 DS |

14. Transposition (Trans):

The antecedent and consequent of a conditional statement may switch places if and only if tildes are inserted before both or tildes are removed from both.

$$(p \supset q) :: (\sim q \supset \sim p)$$

如果昨天我睡得夠，則我今天上課很清醒。
如果我今天上課不是很清醒，則我昨天睡不夠。

15. Material Implication (Impl)

Material implication indicates the truth condition of a conditional statement, which means that a conditional statement is true if and only if the antecedent is false or the consequent is true

$$(p \supset q) :: (\sim p \vee q)$$

如果我口渴，則我喝水。
我不是口渴，或者我喝水。

16. Exportation (Exp)

“If we have both p and q, then we have r”

is logically equivalent to

“If we have p, then if we have q, then we have r.”

$$(p \cdot q) \supset r :: p \supset (q \supset r)$$

如果你有車和有樓時，我們結婚。

如果你有車，則當你有樓時，我們結婚，。

Recall

Rule of replacement

如果你有車和有樓時，則我們結婚。

Exportation

如果你有車，則當你有樓時，我們結婚，。

$$(p \cdot q) \supset r :: p \supset (q \supset r)$$

Consider the following case of using rule of implication

1. $(P \cdot Q) \supset R$

2. $P \supset R$ 1, Simp

1. 如果你有車和有樓時，我們結婚。

2. 如果你有車，我們結婚。 1, Simp

17. Material Equivalent (Equiv):

- (i) The first is the same as the definition of material equivalence given in the previous class.
- (ii) The second formulation indicates the truth condition of “ $p \equiv q$ ”, which means “ $p \equiv q$ ” is *true iff either p and q are both true or p and q are both false*”.

$$(p \equiv q) :: (p \supset q) \cdot (q \supset p)$$

$$(p \equiv q) :: (p \cdot q) \vee (\sim p \cdot \sim q)$$

17. Material Equivalent (Equiv):

我獲得獎學金當且僅當我GPA 4.

(1)我獲得獎學金當我GPA 4，而且當我獲得獎學金則我GPA 4

(2)我獲得獎學金並且我GPA 4，或者我沒有獲得獎學金並且我不是GPA 4.

$$(p \equiv q) :: (p \supset q) \cdot (q \supset p)$$

$$(p \equiv q) :: (p \cdot q) \vee (\sim p \cdot \sim q)$$

18. Tautology (Taut):

To eliminate redundancy in disjunctions and conjunctions.

$$p :: (p \vee p)$$

我好想睡或者我好想睡。

$$p :: (p \cdot p)$$

我好肚餓而且我好肚餓。

Rules of replacement II

14. Transposition (**Trans**)

$$(p \supset q) :: (\sim q \supset \sim p)$$

15. Material Implication (**Impl**)

$$(p \supset q) :: (\sim p \vee q)$$

16. Exportation (**Exp**)

$$(p \cdot q) \supset r :: p \supset (q \supset r)$$

17. Material Equivalent (**Equiv**)

$$(p \equiv q) :: (p \supset q) \cdot (q \supset p)$$

$$(p \equiv q) :: (p \cdot q) \vee (\sim p \cdot \sim q)$$

18. Tautology (**Taut**)

$$p :: (p \vee p)$$

$$p :: (p \cdot p)$$

Discussion 5

1. $S \supset (\sim S \vee P)$ // $S \supset P$
2. $S \supset (S \supset P)$ 1, Impl
3. $(S \cdot S) \supset P$ 2, Exp
4. $S \supset P$ 3, Taut

Discussion 6

1. $P \equiv Q$
2. $Q \supset (R \cdot S)$
3. $\sim S$ // $\sim P$
4. $\sim S \vee \sim R$ 3, Add
5. $\sim R \vee \sim S$ 4, Com
6. $\sim (R \cdot S)$ 4, DM
7. $\sim Q$ 2,5 MT
8. $(P \supset Q) \cdot (Q \supset P)$ 1, Equiv
9. $P \supset Q$ 7, Simp
10. $\sim P$ 6,8, MT

Reference

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